

MENOMINEE REGIONAL, MI, USA

WMO: 726487

Lat: 45.117N

Lon: 87.633W

Elev: 191

StdP: 99.06

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94896

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.2	-19.0	-28.7	0.3	-20.2	-26.0	0.4	-18.2	10.8	-4.1	9.2	-4.9	2.3	280	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	29.9	22.0	28.0	20.7	27.1	19.9	23.3	28.1	22.0	26.9	20.9	25.4	4.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.7	16.8	26.7	20.5	15.5	25.2	19.0	14.1	23.9	70.2	27.9	65.2	26.9	61.5	25.8	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.2	7.4	-25.4	32.7	3.6	2.1	-28.0	34.2	-30.1	35.4	-32.2	36.6	-34.8	38.1		
(5)			-25.4	25.0	2.7	1.5	-27.3	26.1	-28.9	26.9	-30.4	27.7	-32.4	28.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-7.9	-6.2	-1.2	5.0	11.9	17.6	20.4	19.3	15.4	8.9	2.1	-4.3	(6)
(7)		DBStd	10.97	6.19	6.11	5.97	4.53	4.48	4.05	3.31	3.05	4.29	4.59	5.05	5.70	(7)
(8)		HDD10.0	2317	554	454	349	161	29	1	0	0	6	78	241	444	(8)
(9)		HDD18.3	4406	813	687	605	401	206	61	17	24	108	294	487	703	(9)
(10)		CDD10.0	1156	0	0	3	10	89	228	323	289	167	43	4	0	(10)
(11)		CDD18.3	203	0	0	0	0	7	39	82	54	19	2	0	0	(11)
(12)		CDH23.3	1546	0	0	1	4	74	339	656	346	116	9	0	0	(12)
(13)		CDH26.7	365	0	0	0	1	13	90	176	64	20	1	0	0	(13)
(14)	Wind	WSAvg	3.4	3.4	3.5	3.7	4.1	3.7	3.2	2.9	2.8	3.1	3.4	3.4	3.2	(14)
(15)	Precipitation	PrecAvg	763	38	26	43	67	88	98	90	77	81	75	50	40	(15)
(16)		PrecMax	958	138	56	101	119	144	223	164	143	157	181	146	89	(16)
(17)		PrecMin	583	10	4	9	18	35	33	29	13	32	11	6	6	(17)
(18)		PrecStd	91	28	12	23	30	31	57	39	32	35	43	33	23	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	9.2	18.0	22.8	31.9	32.6	31.1	29.1	24.1	17.4	10.6	(19)	
(20)			MCWB	4.0	6.0	12.3	14.1	18.5	23.1	24.0	22.7	22.2	17.8	11.6	8.7	(20)
(21)		2%	DB	3.9	6.1	12.8	17.9	25.7	28.9	30.2	28.0	26.3	21.3	13.6	7.0	(21)
(22)			MCWB	1.6	2.7	8.3	10.9	17.0	21.0	22.3	21.0	19.8	16.4	9.9	4.6	(22)
(23)		5%	DB	2.3	3.4	9.8	15.1	22.7	27.4	28.5	27.1	24.1	18.4	11.4	4.6	(23)
(24)			MCWB	0.8	1.1	5.8	8.7	15.7	19.7	21.6	20.1	18.6	14.2	8.2	2.6	(24)
(25)		10%	DB	1.0	2.1	7.2	12.5	20.0	25.1	27.2	25.2	22.6	16.3	9.0	2.8	(25)
(26)			MCWB	-0.3	0.2	3.6	7.2	13.9	18.3	20.3	19.0	17.8	12.8	6.6	1.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	6.6	14.0	15.2	20.5	24.3	25.4	24.0	23.1	20.3	12.9	9.0	(27)
(28)			MCDWB	6.6	8.5	18.3	22.2	26.2	29.7	31.0	28.5	27.4	22.9	16.5	10.3	(28)
(29)		2%	WB	2.1	3.2	9.1	11.8	18.5	22.0	23.7	22.5	21.2	16.9	10.5	4.9	(29)
(30)			MCDWB	3.6	5.4	12.6	16.7	23.4	28.0	28.5	27.0	25.0	19.8	13.4	6.5	(30)
(31)		5%	WB	0.7	1.4	5.9	9.6	16.4	20.5	22.4	21.2	19.8	14.9	8.4	3.1	(31)
(32)			MCDWB	2.1	3.4	9.4	14.5	21.5	25.8	27.1	25.4	23.4	18.1	10.6	4.5	(32)
(33)		10%	WB	-0.6	0.1	3.7	7.7	14.6	19.1	21.1	20.2	18.6	12.9	7.0	1.5	(33)
(34)			MCDWB	0.7	1.7	6.7	12.0	19.4	23.9	25.9	24.0	22.0	15.8	9.0	2.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.9	8.5	9.2	9.5	10.4	10.2	10.6	9.9	10.1	8.9	7.4	7.1	(35)
(36)			MCDBR	8.0	8.4	12.1	14.4	14.2	12.2	12.0	11.1	11.5	11.7	9.9	7.8	(36)
(37)			MCWBR	6.9	6.6	8.3	9.1	8.1	6.8	6.8	6.3	7.0	7.8	7.4	6.8	(37)
(38)		5% WB	MCDWB	7.8	7.8	11.5	13.5	13.1	11.2	10.9	9.7	10.4	10.2	9.0	7.3	(38)
(39)			MCWBR	7.1	6.4	8.4	9.2	8.2	6.9	6.8	5.9	6.8	7.6	7.3	6.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.260	0.289	0.328	0.358	0.391	0.393	0.384	0.386	0.362	0.328	0.299	0.270	(40)	
(41)		taud	2.461	2.358	2.369	2.360	2.323	2.352	2.394	2.382	2.443	2.529	2.530	2.491	(41)	
(42)		Ebn at Noon	875	898	904	902	878	875	879	864	858	847	817	823	(42)	
(43)		Edh at Noon	68	91	105	116	124	121	115	112	97	78	65	60	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.43	2.43	3.63	4.57	5.54	5.99	6.16	5.23	4.05	2.49	1.48	1.09	(44)	
(45)		RadStd	0.15	0.26	0.34	0.43	0.45	0.46	0.26	0.35	0.30	0.26	0.14	0.11	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page